

Lecture 7

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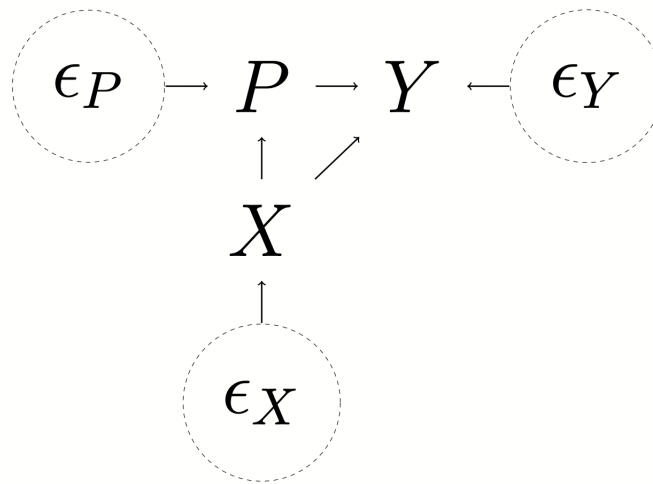
Causal Inference via Directed Acyclical Graphs

Introduction

- The purpose of this module is to provide a more formal and general treatment of acyclic nonlinear (and nonparametric) structural equation models (SEMs) and corresponding causal directed acyclic graphs (CDAGs).
- We discuss the concepts and identification results provided by Judea Pearl and his collaborators and by James H. Robins and his collaborators.
- These concepts allow us to rigorously define causal effects in fully nonlinear models and obtain their nonparametric identification from the structure of the CDAGs alone. Causal effects are defined as hypothetical effects of interventions in systems of equations.
- Whether causal effects derived from SEMs approximate policy or treatment effects in the real world depends to a large extent on the degree to which the posited SEM approximates real phenomena.
- Contextual knowledge is often crucial for making the case that a given structural model represents real phenomena sufficiently well to produce credible estimates of causal effects when using observational data.

From Causal Diagrams to Causal DAGs

- Recall our previous discussion of a model for a household's log-demand for gasoline (Y), which is a function of log-price (P) and household characteristics (X):



The causal DAG for a model of household's log-demand for gasoline.

- Formal causal nonlinear DAGs generalize linear parametric models to general nonparametric forms with the corresponding TSEM:

$$Y := f_Y(P, X, \epsilon_Y),$$

$$P := f_P(X, \epsilon_P),$$

$$X := \epsilon_X$$

where f 's are said to be deterministic structural functions and $\epsilon_Y, \epsilon_P, \epsilon_X$ are structural shocks that are independent of each other.

- When we say that the TSEM is structural, we mean that it is defined by a structure made up of a set of stochastic processes:

$$Y(p, x) := f_Y(p, x, \epsilon_Y),$$

$$P(x) := f_P(x, \epsilon_P),$$

$$X := \epsilon_X,$$

indexed by $(p, x) \in \mathcal{P} \times \mathcal{X}$, called structural functions or structural potential outcome processes.

Moreover, we assume

- *Exogeneity*: Stochastic shocks ϵ_P , ϵ_X , and ϵ_Y are generated as independent variables outside of the model
- *Consistency* The endogenous variables are generated by recursive substitutions:

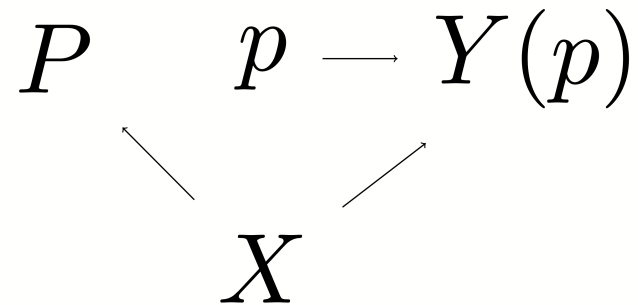
$$Y := Y(P, X), \quad P := P(X), \quad X := \epsilon_X;$$

- *Invariance*: The structure remains invariant to changes of the distribution of stochastic shocks ϵ .

This means that the structure will be assumed to be preserved under various interventions as defined next.

Interventions

- We now consider *fix interventions* that induce single-world intervention graphs (SWIGs).
- The fix intervention was introduced in Heckman and Pinto (2015) and SWIGs were developed by Richardson and Robins (2013).



Causal DAG describing the counterfactual SEM induced by setting $P = p$.

- We split P it into two nodes – P and a deterministic node p – where all the outgoing arrows from P are removed. The fixed node p then inherits the outgoing arrows from the original P .
- The corresponding counterfactual SEM is

$$\begin{bmatrix} Y(p) \\ P \\ X \end{bmatrix} := \left(\begin{bmatrix} Y \\ P \\ X \end{bmatrix} : \text{fix}_Y(P = p) \right) := \begin{bmatrix} f_Y(p, X, \epsilon_Y) \\ P \\ X \end{bmatrix}.$$

- The fix intervention merely says that we are setting $P = p$ in the Y equation.

- The variables $Y(p)$, P , and X are the counterfactuals generated by this intervention.
- The intervention does not affect the P and X equations, nor does it affect ϵ_Y in the Y equation.
- The SWIG allows us to immediately see that conditional exogeneity (ignorability) holds:

$$Y(p) \perp\!\!\!\perp P \mid X,$$

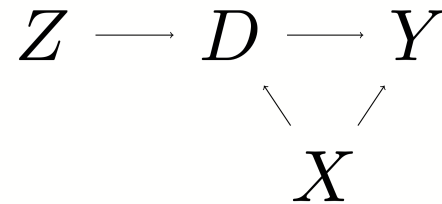
- Therefore we can identify the counterfactual regression $\mathbb{E}[Y(p) \mid X]$ by the “factual” regression $\mathbb{E}[Y \mid P = p, X]$,

$$\mathbb{E}[Y(p) \mid X] = \mathbb{E}[Y(p) \mid P = p, X] = \mathbb{E}[Y \mid P = p, X],$$

invoking conditional independence and consistency arguments.

DAGs and Acyclic SEMs (ASEMs)

- We will give a sequence of formal definitions, but we first begin with examples which introduce these definitions informally.
- A directed acyclic graph (DAG) is a collection of nodes and directed edges with no cycles:



Example: DAG-1

- The system of triangular structural equations corresponding to the example above is

$$Y := f_Y(D, X, \epsilon_Y),$$

$$D := f_D(Z, X, \epsilon_D),$$

$$X := \epsilon_X,$$

$$Z := \epsilon_Z,$$

where ϵ_Y , ϵ_X , ϵ_D , and ϵ_Z are mutually independent.

Consider the DAG-1. Here we can *say* that:

- X is a *parent* of its children D and Y ;
- D and Y are descendants of Z ;
- There is a *directed path* from Z to Y ;
- There are two *paths* from Z to X , but no directed path;
- D is a *collider* of the path $Z \rightarrow D \leftarrow X$;
- D is a *noncollider* of the path $Z \rightarrow D \rightarrow Y$;
- $Y \leftarrow X \rightarrow D$ is the backdoor path from Y to D .
- There are no cycles (there is no directed path that returns to the same node).

General definitions

- A graph $\mathbf{G}$ is an ordered pair (V, E) , where $V = \{1, \dots, J\}$ is a collection of vertices/nodes and E is a matrix of edges $e_{ij} \in \{0, 1\}$ – that is, $E = \{e_{ij} : (i, j) \in V^2\}$.
- Given a collection of random variables $X = (X_j)_{j \in V}$, we *identify* each j with the name “ X_j ” whenever convenient. If the edge (i, j) is present, namely $e_{ij} = 1$, we read it as

” $X_i \rightarrow X_j$ ” or ” X_i is an immediate cause of X_j ”.

- Consider a strict partial order $<$ on V induced by E , where $X_j < X_k$ (we read this as “ X_j is determined before X_k ”) means that either $X_j \rightarrow X_k$ or $X_j \rightarrow X_{v_1} \rightarrow \dots \rightarrow X_{v_m} \rightarrow X_k$ is true for some v_ℓ ’s in V . A partial ordering of V exists if for each j the statement $X_j < X_j$ is not true.
- By the identification above, we can replace names X_ℓ ’s with their indices ℓ ’s.

- The graph $G = (V, E)$ is a DAG if the graph has no cycles, that is, if V is partially ordered by the edge structure E .
- In our example we had vertices $V = \{1, 2, 3, 4\}$ identified with Y, D, X, Z , and the edge set

$$E = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

The partial ordering is $X < D, X < Y, Z < D, D < Y$.

- The *parents* of X_j are the set $Pa_j := \{X_k : X_k \rightarrow X_j\}$.
- The *children* of X_j are the set $Ch_j := \{X_k : X_j \rightarrow X_k\}$.
- The *ancestors* of X_j are the set $An_j := \{X_k : X_k < X_j\}$.
- The *descendants* of X_j are the set $Ds_j := \{X_k : X_k > X_j\}$.
- A directed path is a sequence $X_{v_1} \rightarrow X_{v_2} \rightarrow \dots X_{v_m}$
- a non-directed path is a path, where some arrows (but not all) arrows are replaced by $\leftarrow$.
- A *collider* node is a node X_j such that $\rightarrow X_j \leftarrow$.
- A *backdoor* path from X_l to X_k is an undirected path that starts at X_l and ends with an incoming arrow $\rightarrow X_k$.

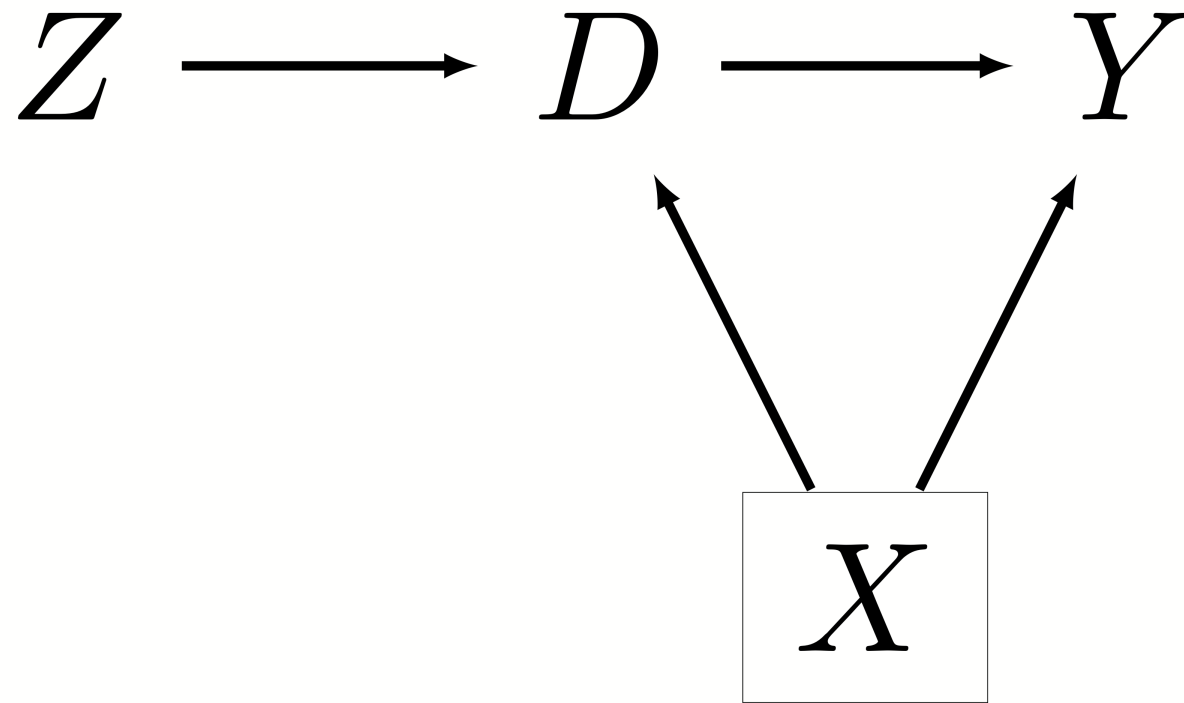
Testable Restrictions by D-Separation

The DAG structure implies another key property: a global Markov condition. In order to define this property, we need a few more definitions.

A path π is said to be blocked by a subset of nodes S if and only if

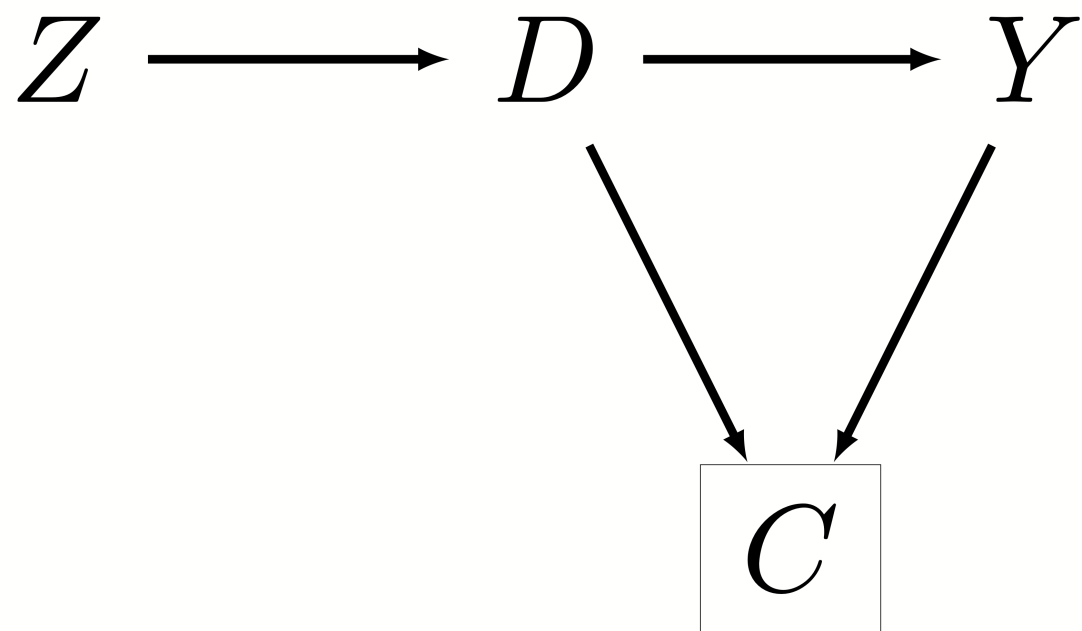
- π contains a chain $i \rightarrow m \rightarrow j$ or a fork $i \leftarrow m \rightarrow j$ such that m is in S ;
- Or, π contains a collider $i \rightarrow m \leftarrow j$, where neither m nor any descendant of m is in S .

A path that is not blocked is called open.



The (backdoor) path $Y \leftarrow X \rightarrow D$ is blocked by conditioning on X .

A path containing a collider is opened by conditioning on it or its descendant:



The path $Y \rightarrow C \leftarrow D$ is blocked, but becomes open by conditioning on C .

- *D-Separation*: Given a DAG G , a set of nodes S d-separates nodes X and Y if nodes in S block all paths between X and Y . D-separation is denoted as

$$(Y \perp\!\!\!\perp_d X \mid S)_G.$$

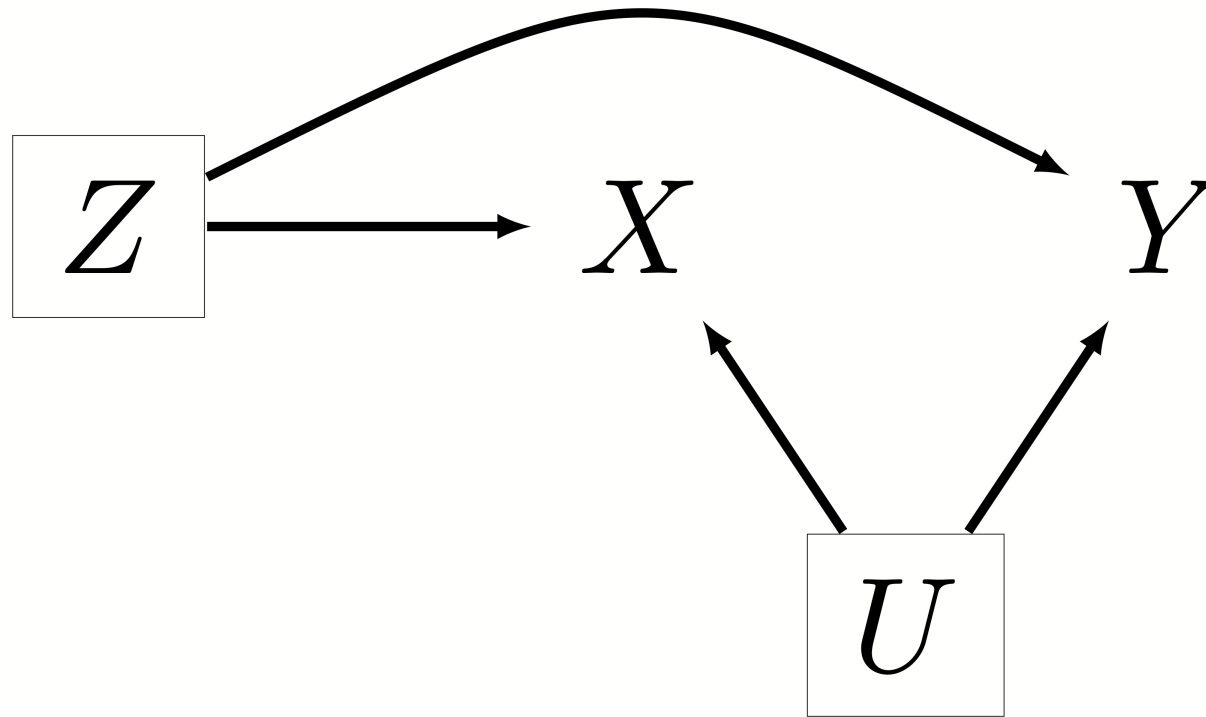
- The following is a deep result concerning the conditional independence relations encoded in the graphs:

Global Markov: d-Separation implies conditional independence:

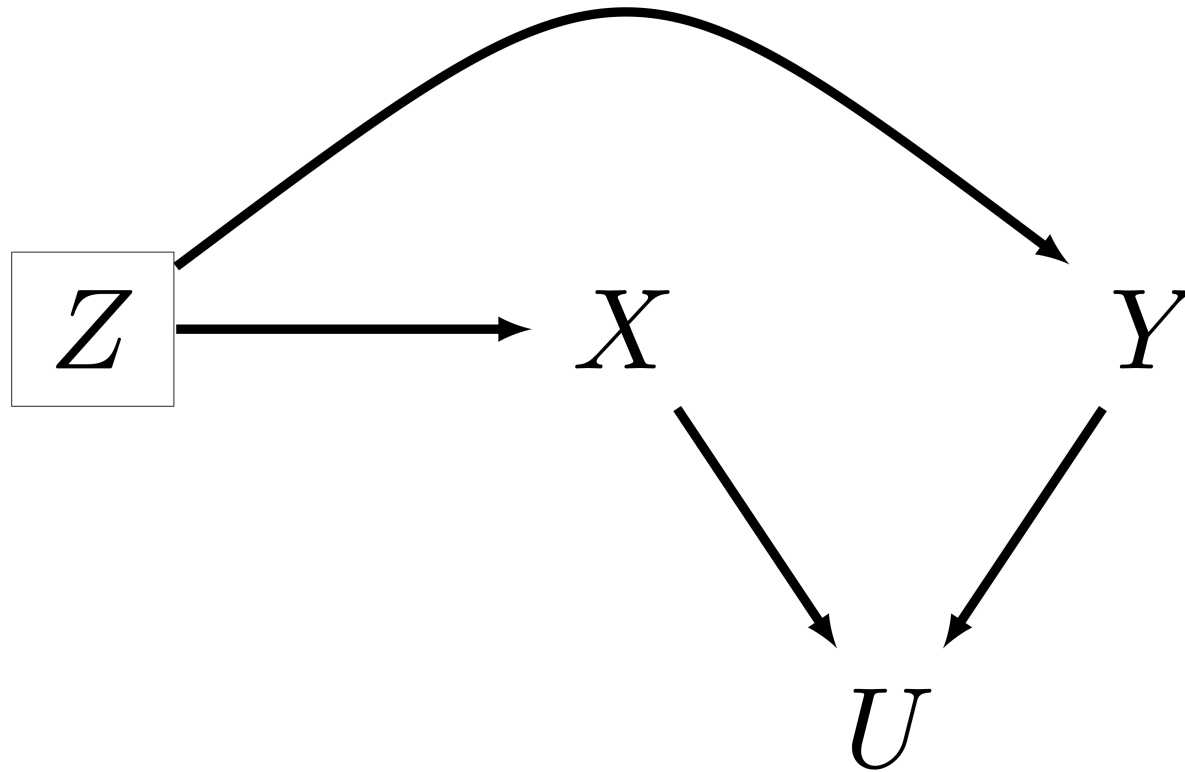
$$(Y \perp\!\!\!\perp_d X \mid S)_G \implies Y \perp\!\!\!\perp X \mid S.$$

- Figuratively speaking, conditioning on S breaks the information flow between Y to X , meaning that Y can't be predicted by X , conditional on S , and vice versa.

- This fundamental result is very intuitive and can be verified directly in simple examples. However, the formal proof is hard (trying to find a simple proof is a good exercise). The reverse implication is not true in general, but is argued to hold “generically”.
- In the following, we show two examples of d -separation.



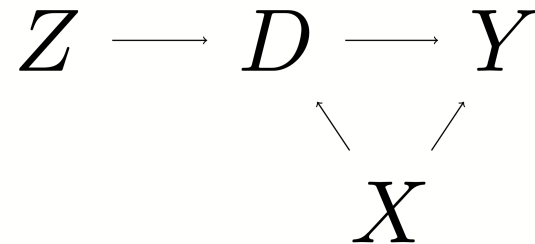
Example of d-separation.



Further example of d-separation.

Counterfactuals Induced by Interventions

- Next, we focus on treatment effect estimation. Again we consider the *DAG-1* given by



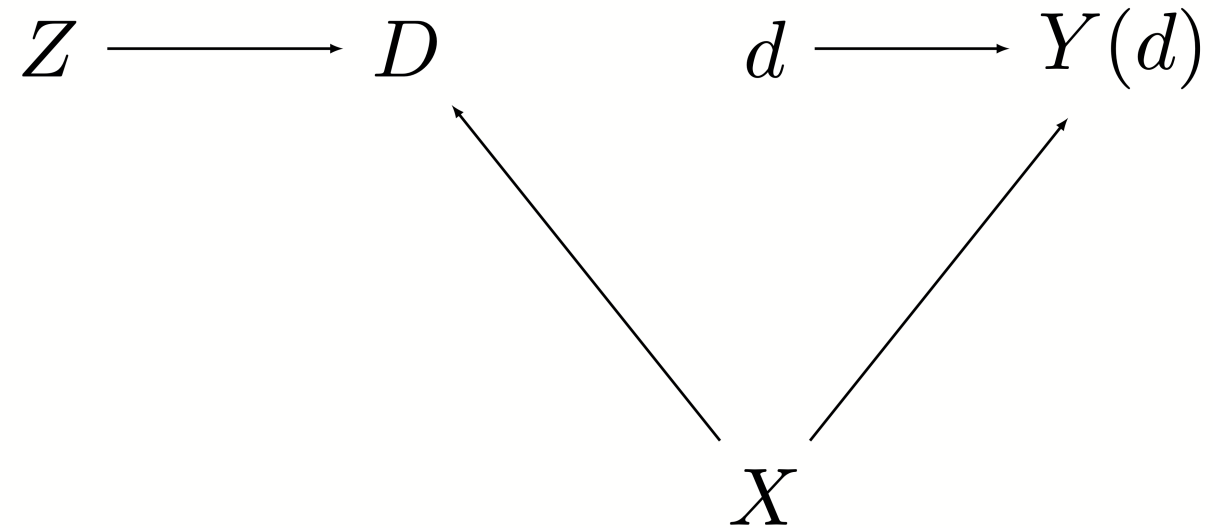
Example: DAG-1

- A counterfactual SEM induced by $\text{fix}(D = d)$ is assumed to take the following form

$$\begin{aligned} Y(d) &:= f_Y(X, d, \epsilon_Y), \\ d, \\ D &:= f_D(X, Z, \epsilon_D), \\ Z &:= \epsilon_Z, \\ X &:= \epsilon_X, \end{aligned}$$

where $\epsilon_X, \epsilon_Z, \epsilon_D, \epsilon_Y$ are mutually independent.

- The corresponding SWIG is given by $\tilde{\mathbf{G}}(d)$:



Single-world intervention graphs (SWIG) induced by the $fix_Y(D = d)$ intervention

Identification by Conditioning

- An adjustment set S is said to be valid for identification of the causal effect of D on Y if the conditional exogeneity/ignorability condition holds

$$Y(d) \perp\!\!\!\perp D \mid S.$$

- In what follows, we present an exhaustive (complete) approach for finding valid adjustment sets by using SWIGs.

- We write down the counterfactual SWIG induced by the

$$\text{fix}(D = d)$$

intervention, which operates on all structural equations defining the descendants of D , by setting $D = d$ in these equations.

- Then, if we have that the potential outcome $Y(d)$ is d -separated from the (policy) variable D by a set of variables S , conditional exogeneity/ignorability holds:

$$Y(d) \perp\!\!\!\perp D \mid S.$$

- In the SWIG graph $DAG-1$, we see that either $S = X$ or $S = (X, Z)$ d-separates $Y(d)$ from D . Therefore either choice of S provides a valid adjustment set for identifying counterfactual predictions.
- Here conditioning on Z is not necessary, though we maintain robustness with respect to the presence of a directed edge from Z to Y in including Z in the conditioning set.

A Complete Criterion for Identification by Conditioning

- Consider any ASEM with DAG G . Let us re-label a policy node X_j as D , and let Y , an outcome of interest, be any other descendant of D .
- Consider a Swig DAG $\tilde{G}(d)$ which is induced by the $\text{fix}(D = d)$ intervention. Consider any other subset of nodes S that appears in both G and $\tilde{G}(d)$, such that

$Y(d)$ is d-separated from D by S in $\tilde{G}(d)$.

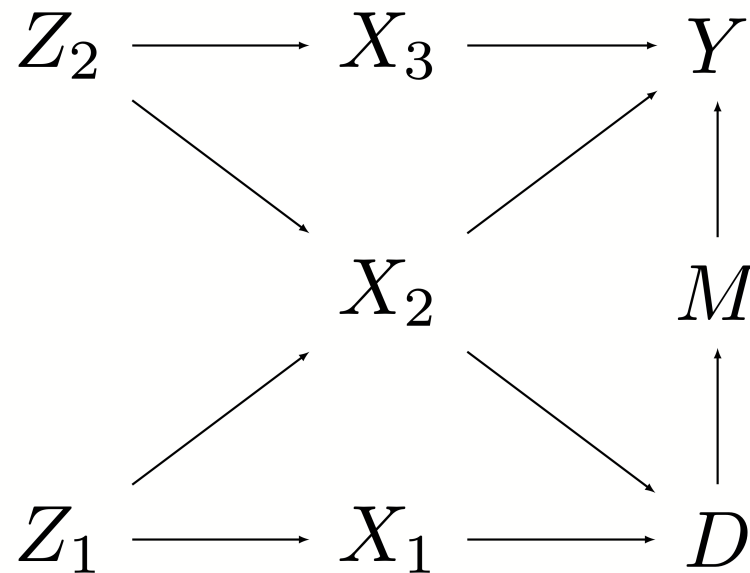
Then, the following conditional exogeneity/ignorability holds

$$Y(d) \perp\!\!\!\perp D \mid S.$$

and $\mathbb{E}[Y(d) | S = s] = \mathbb{E}[Y \mid D = d, S = s]$ holds for all s such that $p(d, s) > 0$.

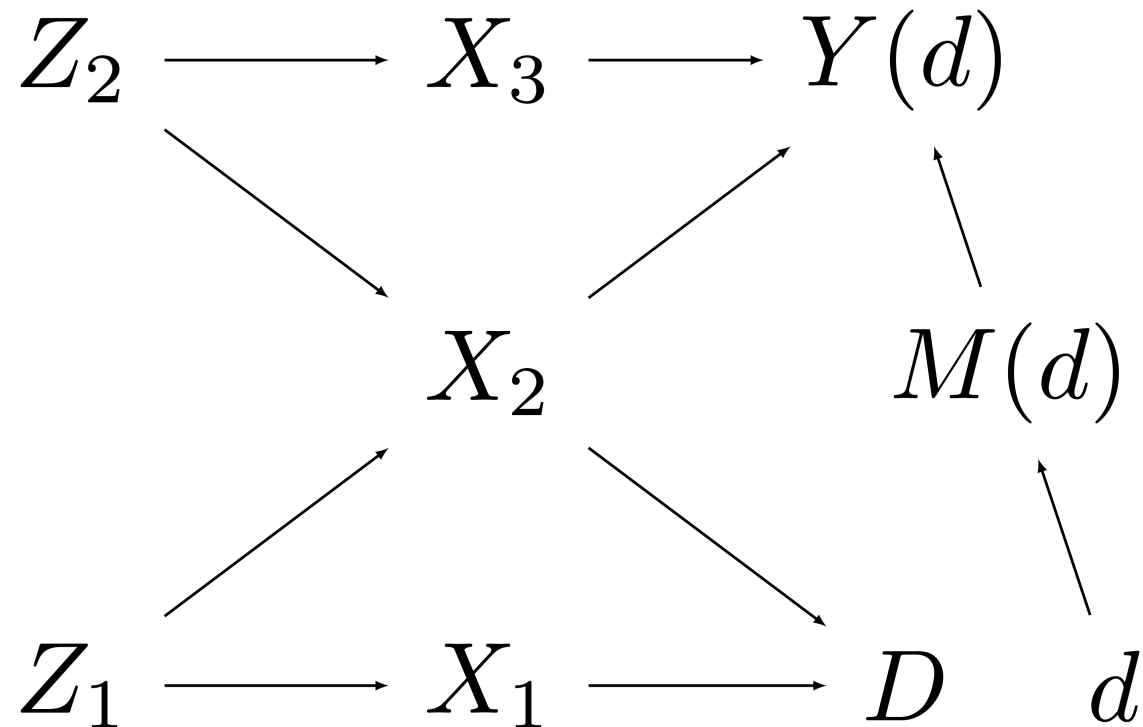
Pearl's Example

- Again, we are interested in the causal effect $D \rightarrow Y$, that is, the effect $d \mapsto Y(d)$.
- We consider the following DAG:



A DAG in “Pearl’s Example”.

- The corresponding SWIG-intervention DAG is shown in following figure:



The DAG induced by the Fix/Swig intervention $\text{fix}(D = d)$ in Pearl's Example.

- In this DAG, valid adjustment sets S include

$$\{X_1, X_2\}, \{X_2, X_3\}, \{X_2, Z_2\}, \{X_2, Z_1\},$$

because each d-separates $Y(d)$ and D by blocking all open paths.

- Conditioning on just X_2 won't work, because it blocks the inner backdoor paths from $Y(d)$ to D , but opens the outer path on which X_2 is a collider.
- To close this opened path it suffices to also condition on X_1 or X_3 or Z_1 or Z_2 .

Useful Adjustment Strategies as Corollaries

- The strategy given above provides a complete (exhaustive) criterion for finding valid adjustment sets.
- We now discuss other frequently used strategies for obtaining valid adjustment sets which are strictly less general.
- Some of these strategies are quite helpful because they are either very simple to apply or can also be used under partial knowledge of the DAG.
- We consider three approaches that allow us to identify the causal effect of D on Y .

1. **Conditioning on all parents** of Y (that are not descendants of D) or of D , or both, is sufficient. This approach is valid, irrespective of the remaining structure of the problem.
2. Conditioning using the **backdoor criterion** enables us to find all minimal adjustment sets.
3. **Conditioning on all common causes** of D and Y is also sufficient.

Conditioning on Parents

- A very simple strategy is conditioning on the parents of Y (that are not descendants of D) or of D , or both.
- In Pearl's example conditioning on parents of D , namely X_1 and X_2 , is sufficient. Alternatively, conditioning on all parents of Y that are non-descendants of D , namely X_2 and X_3 , is also sufficient.
- Here we should not condition on M , because it is a descendant of D .

- Adjusting for parents is a very useful strategy, because it only requires knowledge of parents in a DAG without precise knowledge of the remaining graph structure.
- Conditioning on parents is also behind the propensity score strategies used in many experimental or quasi- experimental empirical analyses.
- If the propensity score is known, it can be used as a parent of D itself.
- Finally, conditioning on parents of Y is most useful for attaining maximal statistical efficiency, but may be less robust than conditioning on both sets of parents under unforeseen deviations from the given graph structure.
- See VanderWeele and Shpitser (2011) for further detailed discussion of robustness of adjusting for both sets of parents.

Conditioning by Backdoor Blocking

- The adjustment set S is valid if the backdoor criterion is satisfied: No element of S is a descendant of D , and all backdoor paths from Y to D are blocked by S .
- In other words, if a collection of random variables S satisfies the backdoor criterion with respect to (D, Y) , then conditioning on S identifies the causal effect of D on Y .
- The basic idea is that if we block the backdoor path, we remove all channels of non-causal association between D and Y .
- Application of the backdoor criterion can produce all minimal adjustment sets.

- In Pearl’s example, the “unintervened” graph has two backdoor paths from D to Y : the inner path

$$D \leftarrow X_2 \rightarrow Y$$

and the outer path

$$D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \leftarrow Z_2 \rightarrow X_3 \rightarrow Y.$$

- Conditioning on just X_2 won’t work, because it blocks the inner backdoor paths from Y to D , but opens the outer path on which X_2 is a collider.
- To close this opened path it suffices to condition on X_1 , X_3 , Z_1 , or Z_2 .
- For example, conditioning sets $S_1 = \{X_1, X_2\}$ or $S_2 = \{X_2, X_3\}$ are valid.
- Conditioning on M is obviously not valid – it is a descendant of D , an intermediate outcome.

Conditioning on All Common Causes of D and Y

- Another simple and widely used adjustment strategy is conditioning on all common causes of the outcome variable of interest and the treatment variable.
- Pearl's Example again, the set of common causes of D and Y is $\{Z_1, Z_2, X_2\}$. This set is a valid adjustment set that differs from the sets ones found using the parental strategy.

- Consider any ASEM. Let S be the intersection of the ancestors of D and Y , called the common causes:

$$S = An_D \cap An_Y.$$

Then S is a valid adjustment set.

- The strategy above is commonly used in empirical work. However, VanderWeele and Shpitser (2011) recommend adjusting for the union S of causes of Y or D (excluding descendants of D) in practice, formally quantifying this strategy as the maximally robust strategy under perturbations of a specified DAG structure that preserves S . This strategy is useful when we don't know the parents of Y or D , but only know that S are their ancestors.

Adjustment for the Union of Causes

- Consider any ASEM. Let S be the union of the ancestors of D and Y that excludes descendants of D :

$$S = An_D \cup An_Y \setminus Ds_D.$$

Then S is a valid adjustment set.

- Application of this criterion gives $\{Z_1, Z_2, X_1, X_2, X_3\}$ as a valid adjustment set. If Z_1 and Z_2 are not observed, then adjusting for $\{X_1, X_2, X_3\}$ suffices.

Faithfulness and Causal Discovery

- d-Separation implies conditional independence, but the reverse implication

$$Y \perp\!\!\!\perp X|S \implies (Y \perp\!\!\!\perp_d X|S)_G$$

is not true in general.

- If we restrict attention to the set of distributions p of random variables associated with graph G such that the reverse implication holds, we are said to impose the *faithfulness* assumption on p .

Notes

Any econometric study that relies on conditioning to learn the causal effect must have a thought process that justifies this approach. The DAG/ASEM framework is a rigorous form of this process, which enables explicit incorporation of domain knowledge, automatic checking of identifiability, and automatic deduction of testable restrictions. Graphs also offer an effective way of communicating models.

Study Problems

1. For Pearl's example, write out the parents, non-parents, descendants, and non-descendants of nodes X_2 and M . List all the backdoor paths between Y and X_2 . Can you identify the effect of X_2 on Y by conditioning?

2. Consider Pearl's example and answer the following questions.

- What are the testable implications of the assumptions embedded in the model? Hint: The testable implications are derived from the d-separation criterion.
- Assume that only variables D , Y , X_2 and X_3 are measured, are there any testable implications?
- Now assume only D , Y , and X_2 are measured. Are there any testable implications?
- Now assume that all of the variables but X_3 (8 in total) are measured. Are there any testable restrictions?
- Assume that an alternative model, has the same structure, but with the $X_2 \rightarrow D$ arrow reversed. What statistical test would distinguish between the two models?

3. Add an arrow $Z_2 \rightarrow Z_1$ in Pearl's example and figure out how to identify the effect of $D \rightarrow Y$ by conditioning, of $D \rightarrow M$ by conditioning, and of $M \rightarrow Y$ by conditioning (Note that valid conditioning sets may be empty). Can you identify the effect of $X_2 \rightarrow Y$? If so, how? You may solve this analytically or using any of the causal identification packages.

4. Add an arrow $X_1 \rightarrow M$ in Pearl's example and figure out how to identify the effect of $D \rightarrow Y$ by conditioning, of $D \rightarrow M$ by conditioning, and of $M \rightarrow Y$ by conditioning. Can you identify the effect of $X_2 \rightarrow Y$? If so, how? You may solve this analytically or using any of the causal identification packages.

Additional resources

- [Dagitty.Net](#) is an excellent online resource where you can plot and analyze causal DAG models online. It contains many interesting examples of DAGs used in empirical analysis in various fields.
- [Causalfusion.Net](#) is another excellent online resource where you can plot and analyze causal DAG models. This resource covers many different deviations from the standard framework.

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